

Abstract. The service integration of transmission and distribution network is a trend in regional power grid dispatch operation and development. In consideration of the present energy management system (EMS) and distribution management system (DMS) development in an independent mode, the service demand of transmission and distribution network in coordinated operation and analytic calculation is analyzed. The coordinated operation overall structure of transmission and distribution network advanced application are proposed. On this basis, several key technologies, including unified modeling, topology analysis, power flow calculation, risk assessment and fault handling in transmission and distribution network are treated, while references for coordinated operation and system construction in transmission and distribution network dispatch service are provided.

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Key words: district power grid; dispatching control system; transmission grid; distribution grid; coordinated operation; risk assessment